Yaoteng Tan

Ph.D. Candidate, Electrical and Computer Engineering, University of California, Riverside
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RESEARCH INTERESTS

Trustworthy AI, foundation models, machine unlearning, adversarial robustness, vision-language models, text-to-image generation, AI safety, and optimization for machine learning.

EDUCATION

University of California, Riverside Sep 2022 – Jun 2027 expected
Ph.D. Candidate in Electrical and Computer Engineering *Riverside, CA*
Advisor: Prof. M. Salman Asif

Huazhong University of Science and Technology Sep 2018 – Jun 2022
B.S. in Electrical Engineering, Graduated with Honors *Wuhan, China*

PUBLICATIONS

1. **Y. Tan***, Z. Cai*, M. S. Asif. *Targeted Unlearning with Single Layer Unlearning Gradient.* Proceedings of the **International Conference on Machine Learning (ICML)**, 2025. [arXiv] [project page]
2. **Y. Tan***, Z. Cai*, M. S. Asif. *Ensemble-based Blackbox Attacks on Dense Prediction.* Proceedings of the **IEEE Conference on Computer Vision and Pattern Recognition (CVPR)**, 2023. [arXiv] [code]
3. **Y. Tan**, Z. Cai, M. S. Asif. *Modular Energy Steering for Safe Text-to-Image Generation with Foundation Models.* Under review at ECCV 2026. [arXiv]
4. **Y. Tan**, Z. Cai, M. S. Asif. *Transform-dependent Adversarial Attacks.* Under review at TMLR [arXiv]

* Equal contribution.

RESEARCH EXPERIENCE

Graduate Research Assistant Oct 2022 – Present
University of California, Riverside *Riverside, CA*

- Conduct research on trustworthy AI, with emphasis on machine unlearning, adversarial robustness, and safety mechanisms for foundation models.
- Develop efficient algorithms for vision-language models and text-to-image generative models, targeting both empirical performance and mechanistic understanding.
- Design and evaluate methods across modern machine learning benchmarks, including adversarial attack, unlearning, and generative safety settings.

Research Engineer Intern Jun 2026 – Sep 2026 expected
Center for AI Safety *San Francisco, CA*
Mentor: Dr. Steven Basart

- Contributing to AI safety projects that are endeavour for societal impact, open to public, targeting publication in major machine learning venues.

Machine Learning Engineer Intern Jul 2021 – Aug 2021
CloudMinds Technology Inc. *Beijing, China*

- Developed a real-time robotics vision module using object detection and deployed it in an Unreal Engine simulation environment for testing.

SELECTED RESEARCH PROJECTS

Modular Energy Steering for Safe Text-to-Image Generation Aug 2025 – Mar 2026
arXiv:2604.02265 — *Keywords: AI safety, scalable oversight, Text-to-image models*

- Proposed a training-free, and dataset-free inference-time safety steering framework that leverages

pretrained foundation models as oversight signals for text-to-image generation.
- Demonstrated compatibility with diffusion and flow-matching models, including scalable steering for multiple target concepts.
- Reduced average NSFW attack success rate from 82.7% to 8.0% across four red-teaming benchmarks on SD-v1.4, while preserving image quality on general concepts.

Targeted Unlearning with Single Layer Unlearning Gradient Jun 2024 – Jul 2025
ICML 2025 — *Keywords: machine unlearning, post-training adaptation, vision-language models*
- Proposed SLUG, a one-step gradient-based targeted unlearning for multimodal foundation models.
- Achieved 20–50× higher efficiency and improved retention-forgetting trade-off over existing baselines.
- First showed that targeted unlearning can be performed as a layer-wise modular update, enabling localized concept removal while preserving retained knowledge.

Discovery of Sparsity-Constrained Optimization Algorithms Sep 2024 – Jul 2025
GitHub repository — *Keywords: optimization, meta-learning, AI for science*
- Developed a meta-optimization framework for discovering adaptive solvers for sparsity-constrained inverse problems.
- Connected classical optimization structure with data-driven algorithm design for sparse recovery.

Transform-dependent Adversarial Attacks Sep 2022 – Jun 2023
arXiv:2406.08443 — *Keywords: adversarial attacks, physical robustness, imaging pipeline*
- Introduced transform-dependent adversarial attacks that reveal vulnerabilities beyond conventional static perturbation settings.
- Built a camera pipeline that enables optimization through hardware-related image transformations.

Ensemble-based Blackbox Attacks on Dense Prediction Sep 2022 – Jun 2023
CVPR 2023 — *Keywords: black-box attacks, adversarial robustness, dense prediction*
- Proposed a query-efficient black-box attack that aggregates surrogate gradients from model ensemble.
- Achieved high transferability and query-efficiency across detection and segmentation models.

Adversarial Robustness of Computational Imaging Systems Feb 2022 – Jul 2022
Keywords: computational imaging, adversarial robustness, inverse problems
- Evaluated the vulnerability of deep computational imaging models to adversarial perturbations.
- Compared learned imaging pipelines robustness against conventional non-learning methods and explored mitigation strategies.

ACADEMIC SERVICE

Conference Reviewer:
- **2026:** ICLR, CVPR, ECCV, ICIP, WACV, NeurIPS, IEEE Asilomar.
- **2025:** ICCV, ICIP, IEEE Asilomar.
- **2024:** WACV.

Teaching Assistant:
- UCR EE240 Pattern Recognition, Spring 2023 and Spring 2024.
- UCR CS171/EE142 Introduction to Machine Learning and Data Mining, Fall 2023 and Winter 2026.

TECHNICAL SKILLS

Research Areas: Trustworthy AI, safety steering, model-oversight, machine unlearning, adversarial robustness, safety optimization, vision-language models, text-to-image generation.
Programming (ranked by proficiency): Python, PyTorch, MATLAB, C/C++, HTML.
Tools: Git/GitHub, Linux/Unix, LaTeX, Shell, SQL, Qt.
Communication: Mandarin, English, Español.